
# Phase 1: Strategic Domain Selection

Week 4 Update

Naina Misra | Lillian Lam | Chaohui Xu | Yang Cao

# Outline

- Objective & Pipeline
- Recap
- Experimental Results
- Next Steps

# Objective

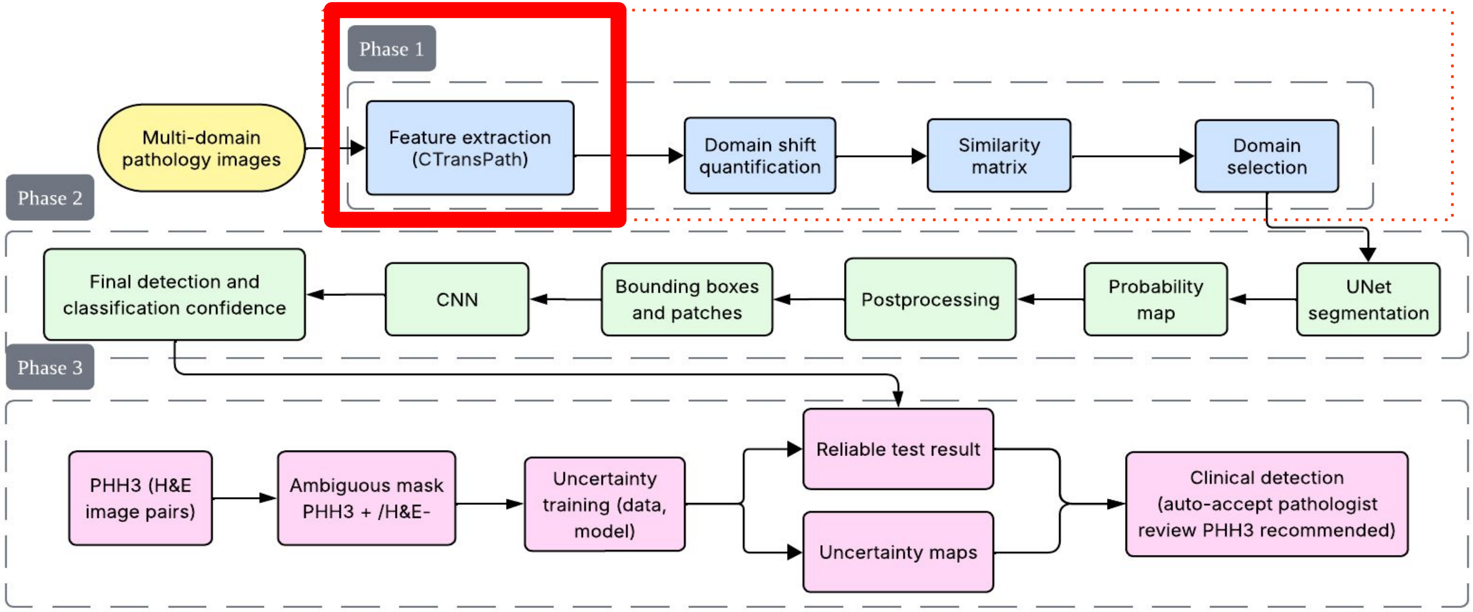
To develop a systematic pipeline to analyze domain relationships, select optimal training domains, and build a robust, ambiguity-aware detection model.

📌 Phase 1: Quantitative Domain Analysis & Strategic Selection

📌 Phase 2: U-Net segmentation & Classification Pipeline

📌 Phase 3: PHH3 Ambiguity Modeling & Uncertainty Quantification

# Pipeline


Phase 1
Multi-domain pathology images
Feature extraction (CTransPath)
Domain shift quantification
Similarity matrix
Domain selection
Phase 2
UNet segmentation
Probability map
Postprocessing
Bounding boxes and patches
CNN
Final detection and classification confidence
Phase 3
PHH3 (H&E image pairs)
Ambiguous mask PHH3 + /H&E-
Uncertainty training (data, model)
Reliable test result
Uncertainty maps
Clinical detection (auto-accept pathologist review PHH3 recommended)

# Outline

- Objective & Pipeline
- Recap
- Experimental Results
- Next Steps

# Recap

Feature Extraction using CTransPath
Patch Extractions of WSIs ⟶ CTransPath feature vectors ⟶ UMAP visualization

# Recap (Next Steps)

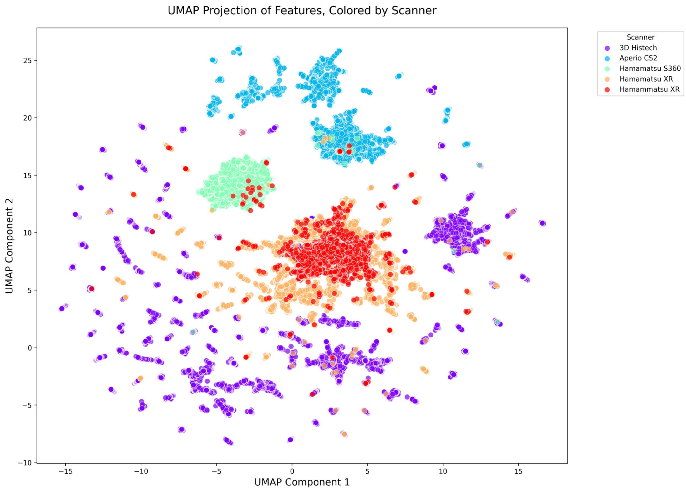
Step 1:
- Normalize the slides to flatten distribution
- Redo the CTransPath and UMAP to see if clusters are closer together
  - see if scanner and lab origin clusters reduce in distance
  - better visualize morphological features

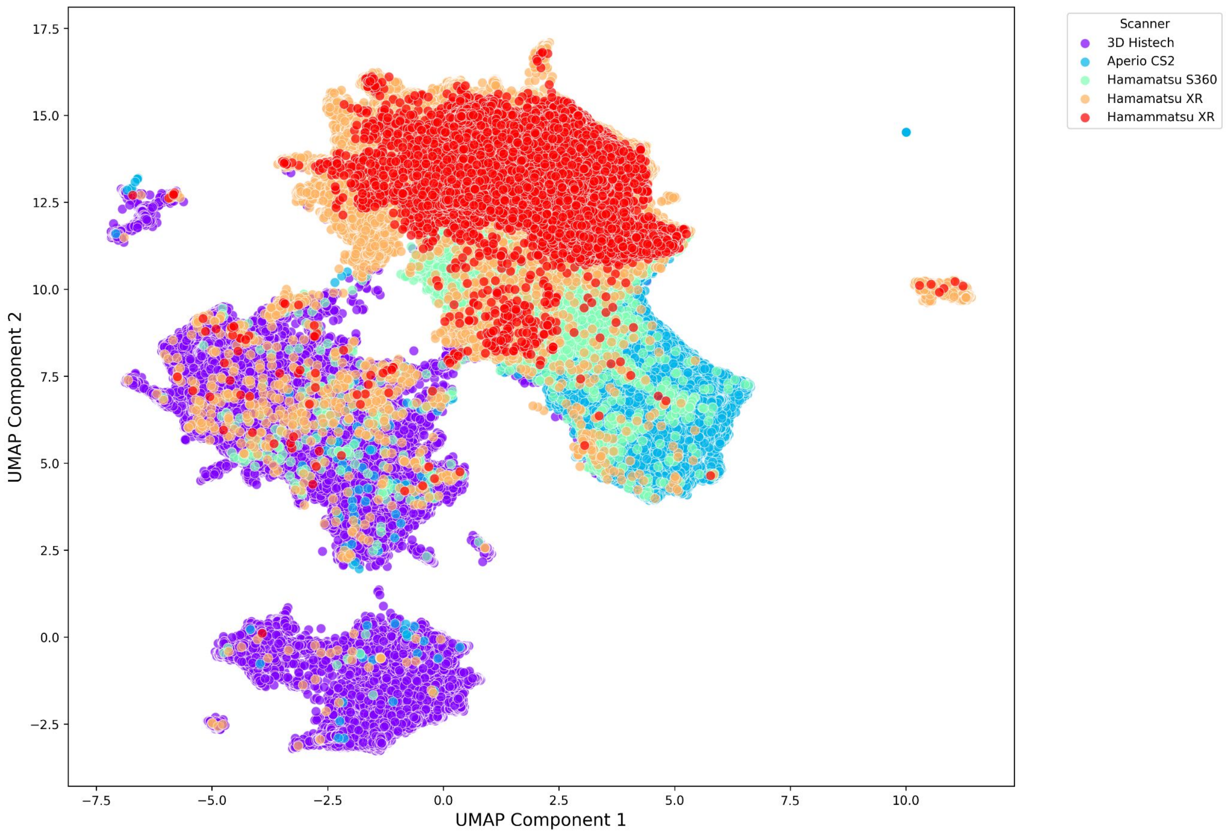
Step 2:
- Feature extraction on unbiased model (EfficientNet)

# Outline

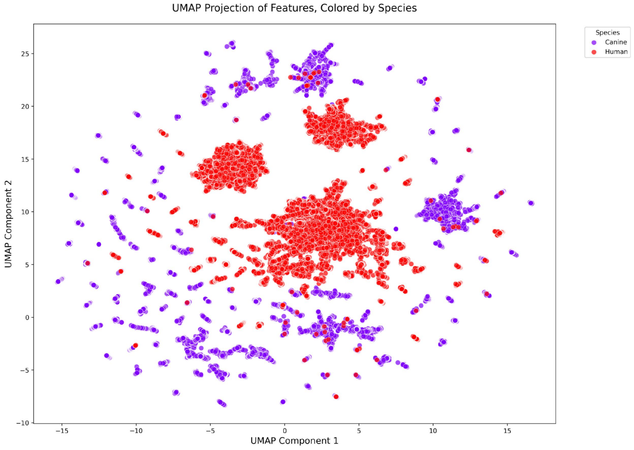
- Objective & Pipeline
- Recap
- Experimental Results (Efficient Net B0)
- Next Steps

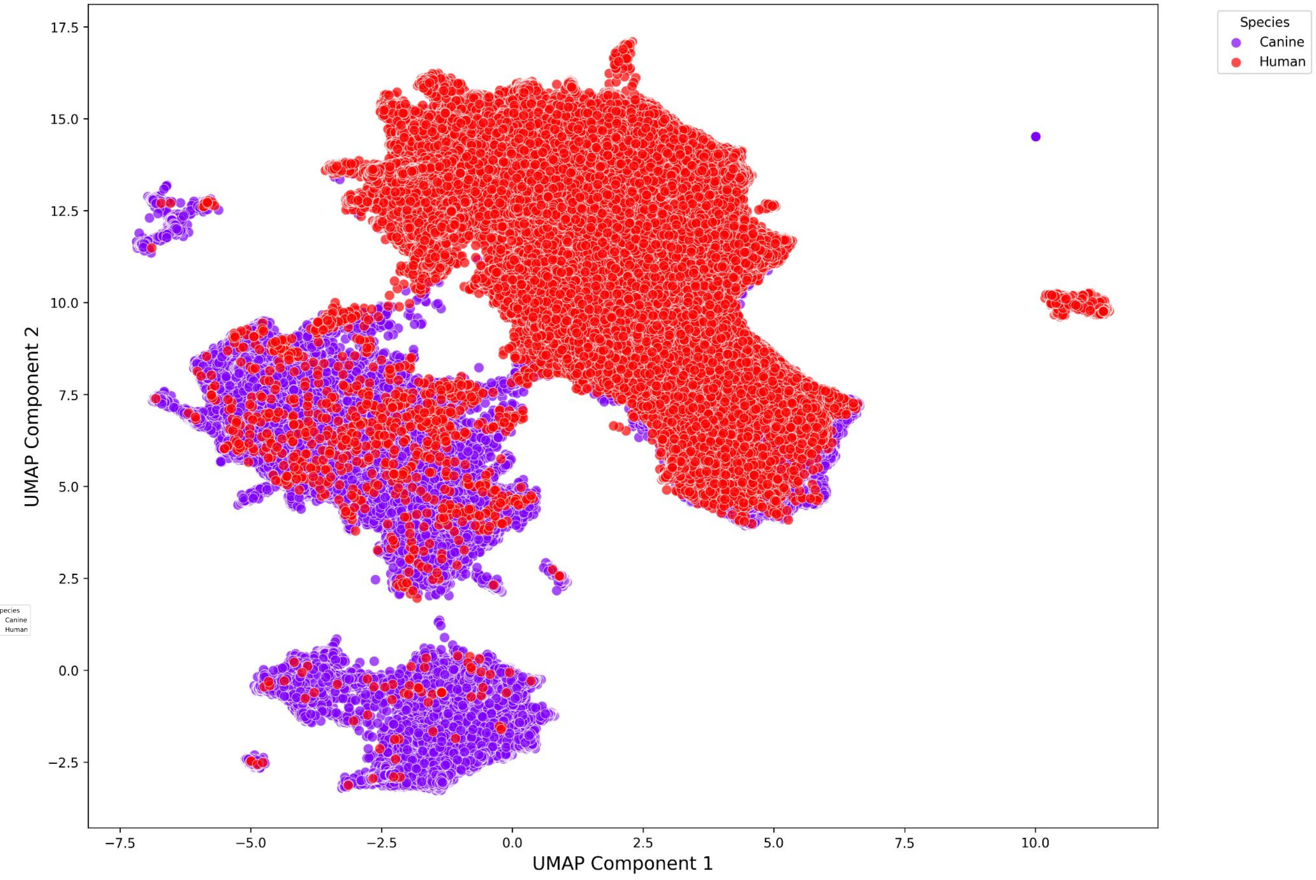
# UMAP: Scanner CTransPath vs Efficient Net B0



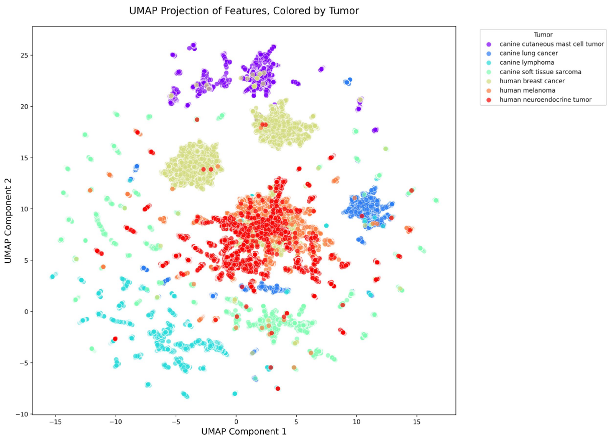



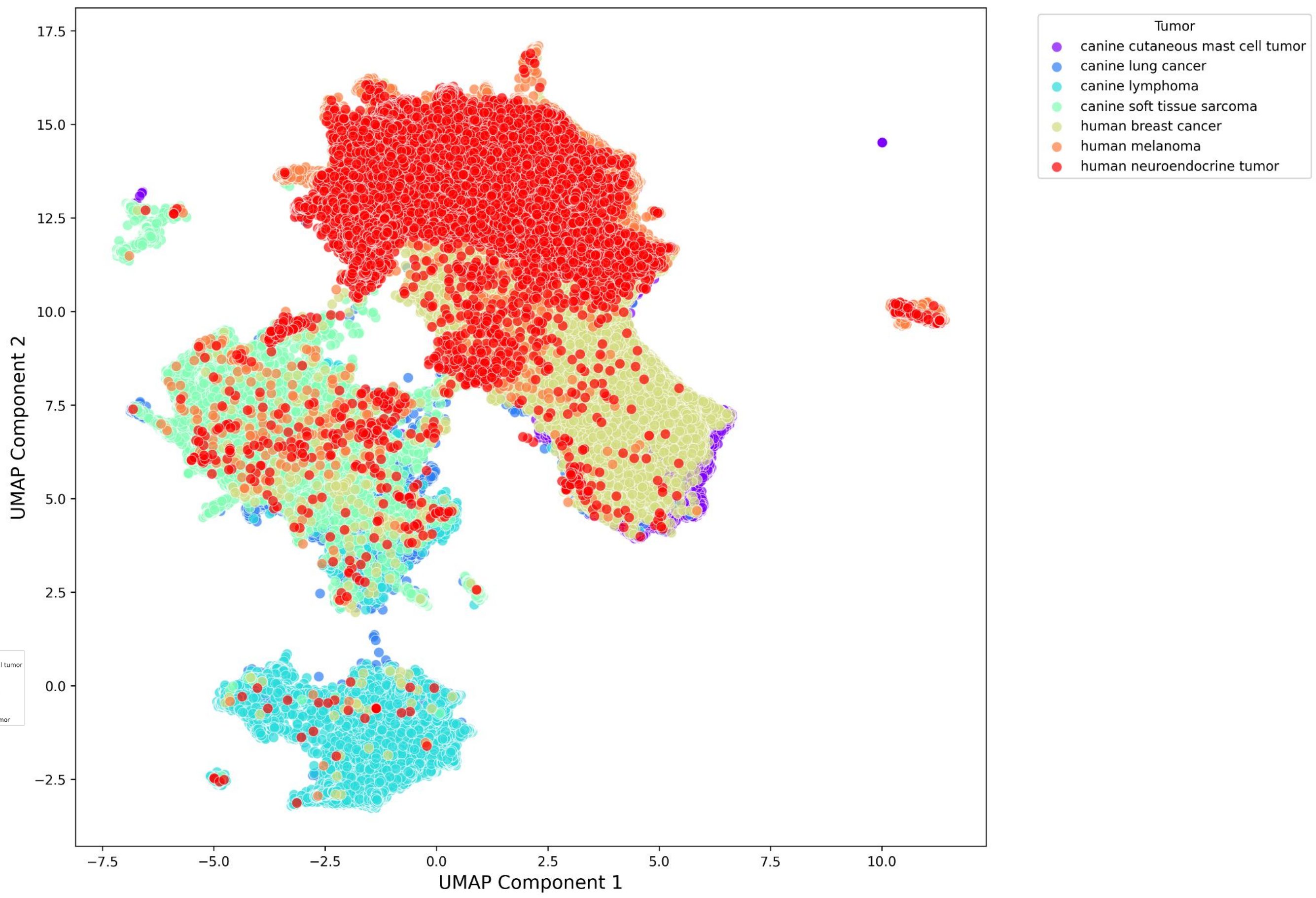
# UMAP: Species CTransPath vs Efficient Net B0



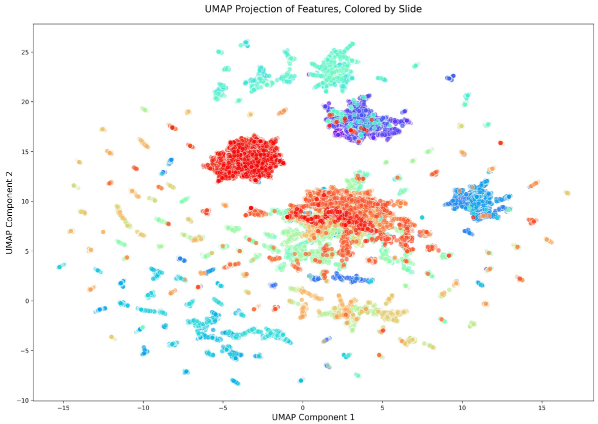



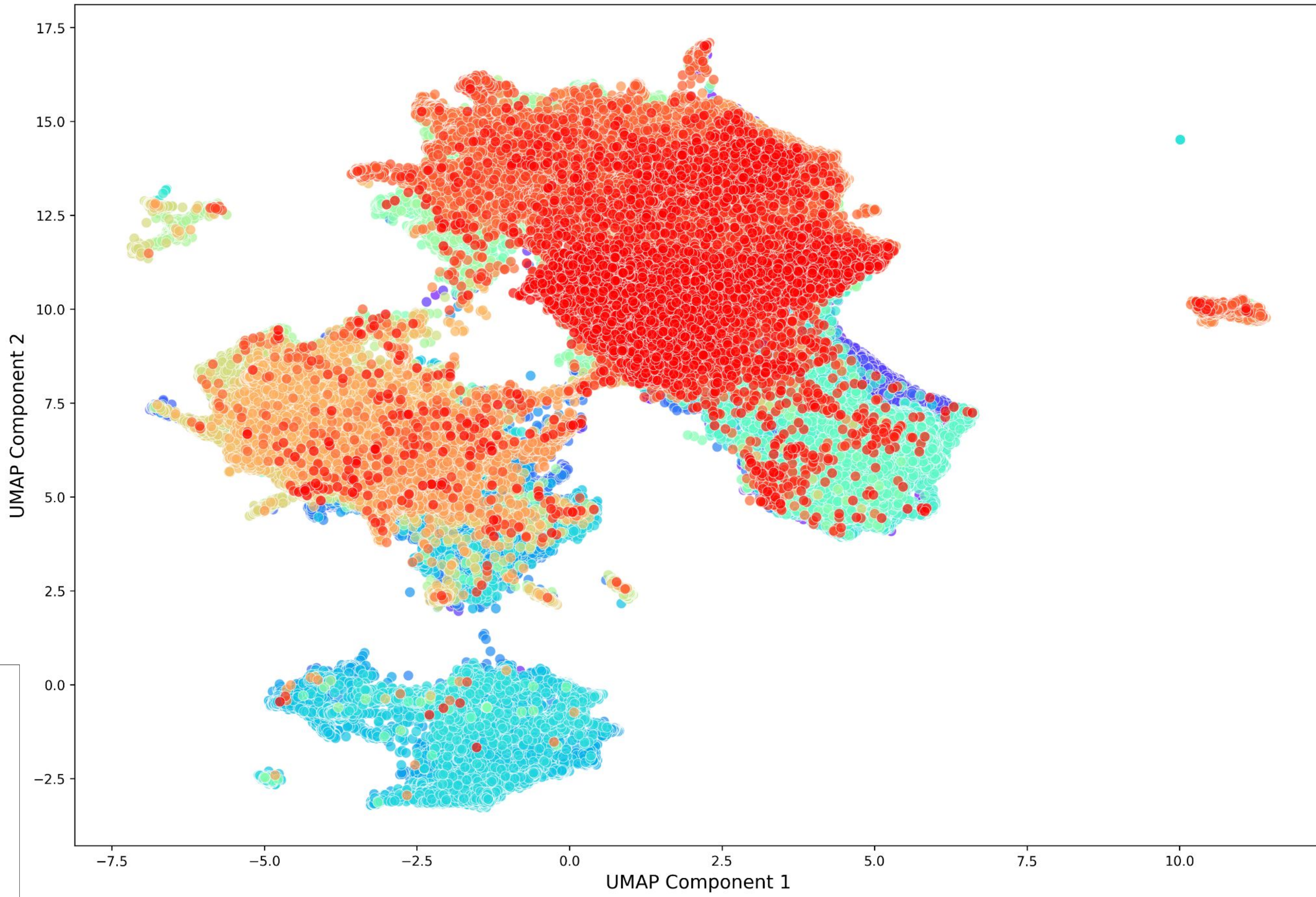
# UMAP: Tumor Type CTransPath vs Efficient Net B0



# UMAP: WSI CTransPath vs Efficient Net Bo

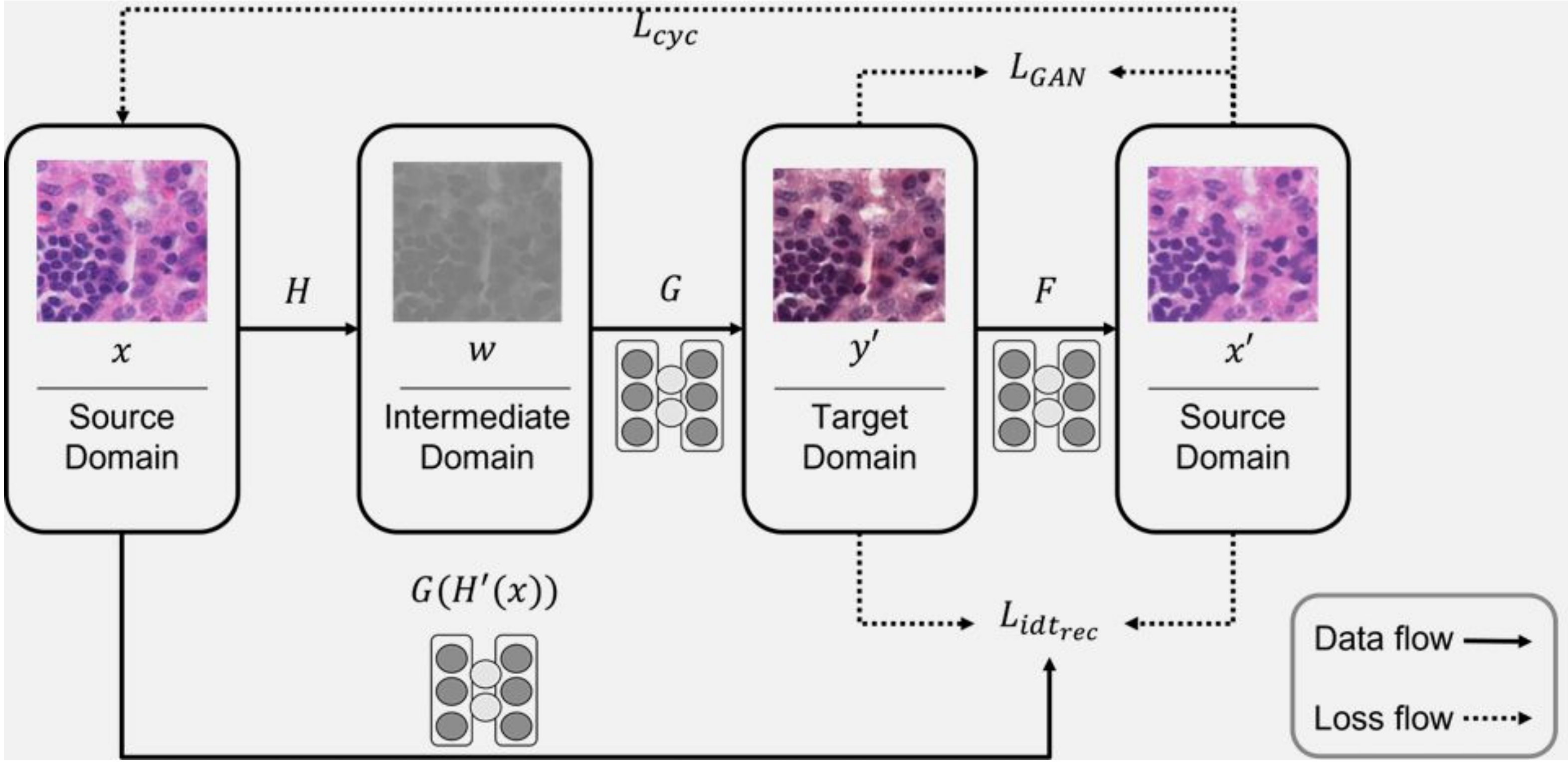


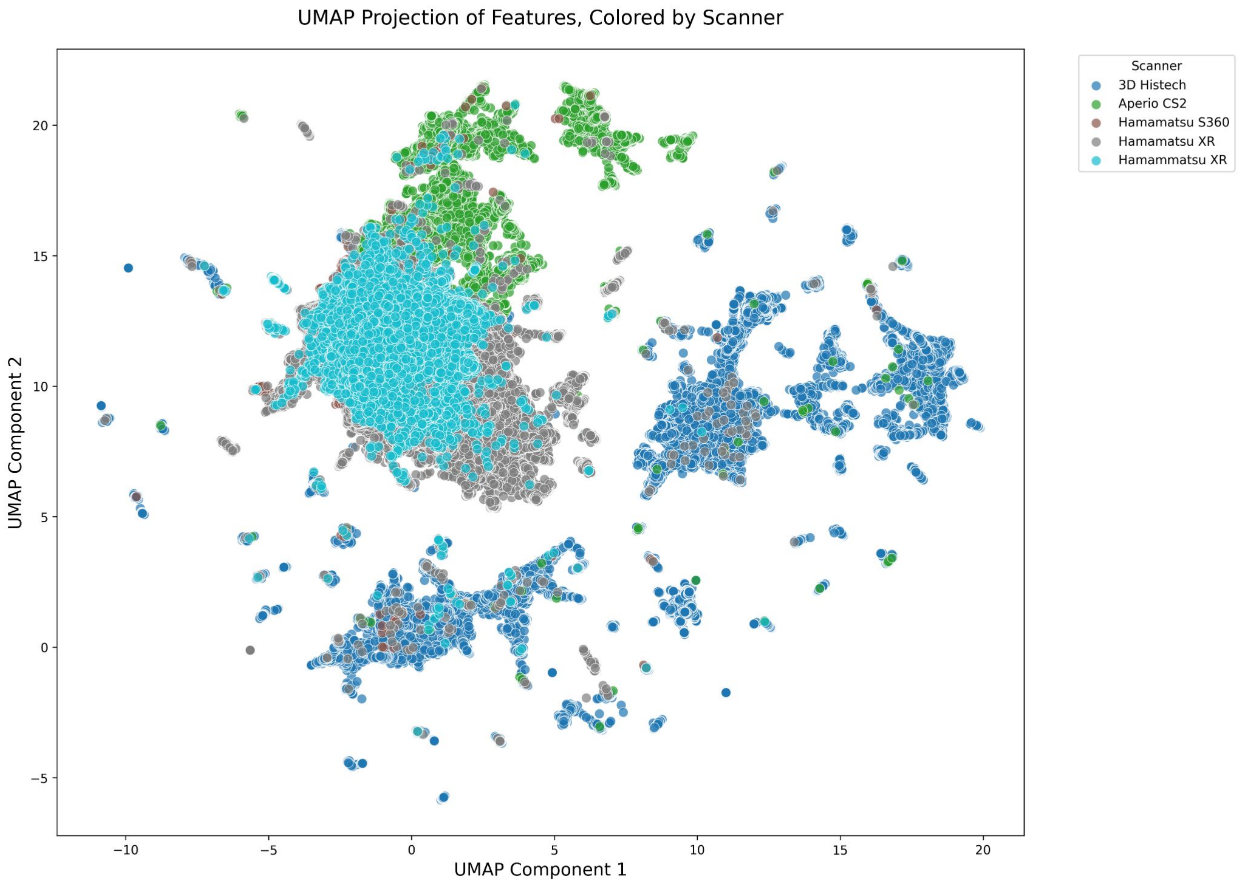
# Outline

- Objective & Pipeline
- Recap
- Experimental Results (Cycle GAN Normalization + CTransPath)
- Next Steps

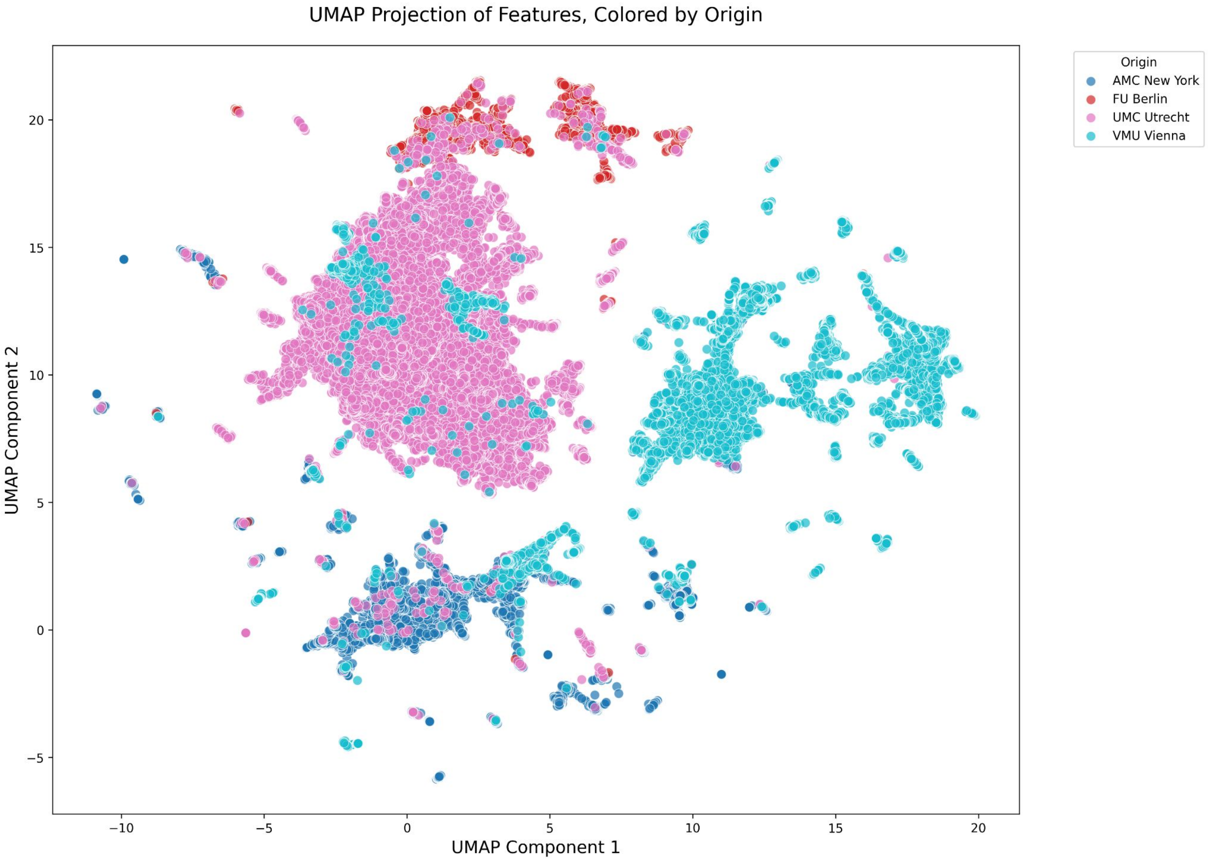
# Cycle GAN (Generative Adversarial Network) Stain Normalization



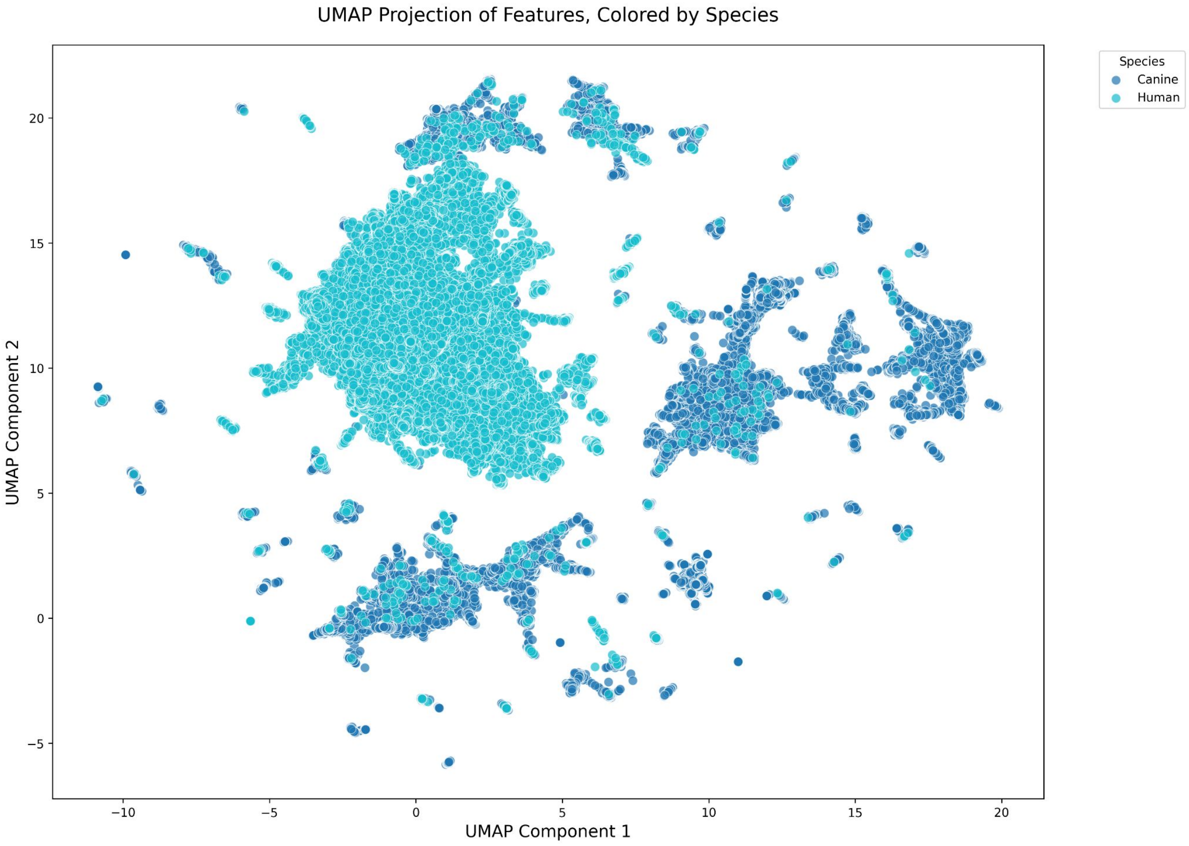
# UMAP: Scanner (Cycle GAN + CTransPath)



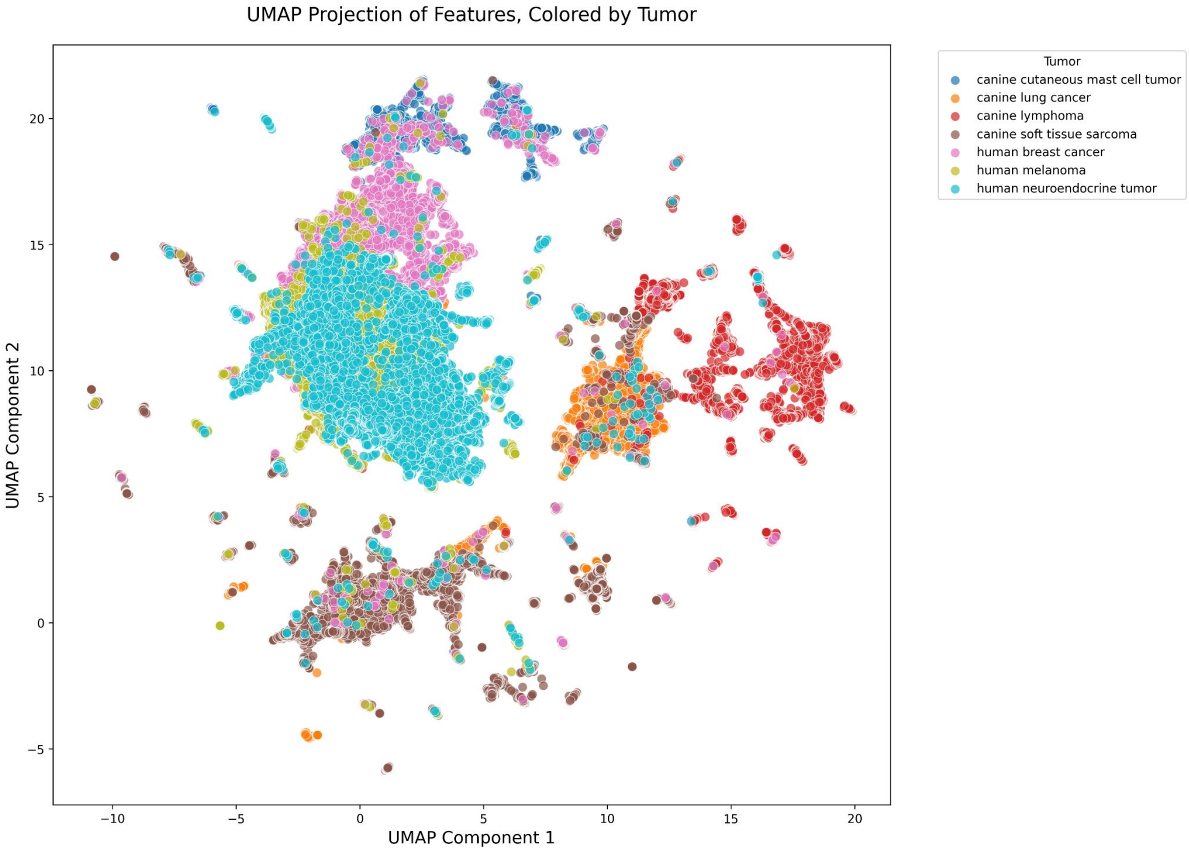
# UMAP: Lab Origin  (Cycle GAN + CTransPath)



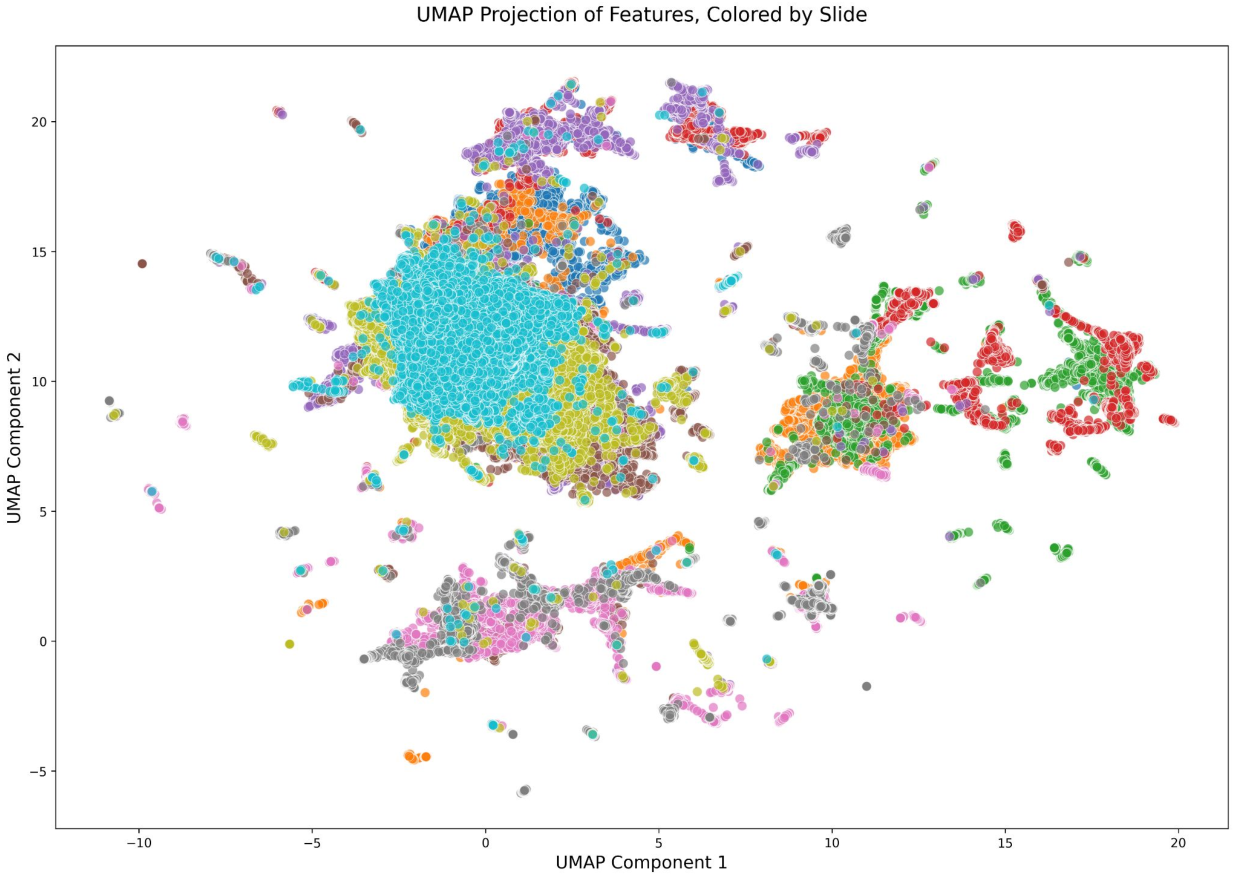
# UMAP: Species (Cycle GAN + CTransPath)



# UMAP: Tumor Type (Cycle GAN + CTransPath)



# UMAP: WSI (Cycle GAN + CTransPath)



# Outline

- Objective & Pipeline
- Recap
- Experimental Results (Cycle GAN Normalization + CTransPath)
- Next Steps

# Next Steps (Upcoming 2-4 Weeks)

Step 1:

- Work with other feature extractors to see if we get better results (w & w/o stain normalization)
  - DINOv3
  - Mask autoencoder
- play around with UMAP parameters
- Determine if there are natural clusters within the feature space (potential areas to start training data selection)
- Reaching out for PHH3 ground-truth dataset

Step 2:

- Domain shift quantification
  - determine an algorithm for weights on each measurement parameter

# Thank you!